SmartSpend — Financial Health Score (FHS) Formula

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Overview

The FHS is a **Prototype Observed Financial Health Indicator** scored 0–100, computed entirely from the user's logged transaction data (no surveys, no self-report). The design adapts the equal-weight, transaction-based structure of the *CBA–Melbourne Institute Observed Financial Health Indicator* (Comerton-Forde, de New, Ribar, Salamanca, Nicastro, & Ross, 2018, Technical Report No. 1).

FHS = S + O + B + L (maximum = 100 points)

Each of the four components is worth **25 points**. Equal weighting follows Melbourne Institute precedent and the OECD/JRC *Handbook on Constructing Composite Indicators* (2008), which establishes equal weighting as the methodologically defensible default when longitudinal data for differential weights is unavailable.

Full Mode — Used when monthly income is set

Component 1 — Savings Rate (S)

```
S = 25 × min(1, savingsRate ÷ 0.20)

where: savingsRate = (monthlyIncome − totalSpent) ÷ monthlyIncome

No-income fallback: S = 20 if totalSpent < ₱5,000                    S =
15 if totalSpent < ₱10,000                    S = 10 otherwise
```

Basis: The 20% savings threshold is from Warren & Tyagi's (2004) 50/30/20 rule, which allocates 20% of net income to savings as the standard personal finance target. Full score is awarded when the user meets or exceeds this benchmark.

Component 2 — Overspend Control (O)

```
O = 25 × (1 − weightedOverDays ÷ activeDays)

where: weightedOverDays = hardOverDays + (softOverDays × 0.5)
       hardOverDays = days of habitual/daily overspending (weight: 1.0×)
       softOverDays = days with one large single-item overspend (weight:
0.5×)
```

Basis: Kahneman & Tversky's (1979) Prospect Theory — losses register approximately twice as strongly as equivalent gains, making overspend penalties motivationally salient. The soft/hard distinction (Thaler & Sunstein, 2008) prevents penalizing a one-time tuition or medical payment the same as daily habitual overspending. The 0.5× coefficient is a calibrated design constant.

Component 3 — Budget Adherence (B)

```
B = 25 × (onBudgetCategories ÷ totalBudgetCategories)

Special case: if no budgets are configured → B = 25 (auto-award)
```

Basis: Financial Health Network's FinHealth Score® (2021) Plan pillar identifies budget adherence as a core financial health behavior. Auto-award is justified by BSP CFIS (2025) data showing fewer than 50% of Filipino households maintain any written budget — the system does not penalize users for lack of initial setup.

Component 4 — Logging Consistency (L)

```
L = 25 × (loggedDays ÷ activeDays)
```

Basis: Deci & Ryan's (2000) Self-Determination Theory (SDT), Competence dimension — consistent, intrinsically motivated behavior drives long-term habit formation. Ariely (2008) and Duhigg (2012) further support that consistency in financial tracking is more predictive of good outcomes than volume on a single day.

Adjustments (applied after summing S + O + B + L)

Warning Decay

```
penalty = min(decayDays, 3) × 5   → subtracted from FHS (max −15 pts)

Triggers when: user continues spending in a budget category after an
alert
Exempt categories: Bills, Health, Education
```

Basis: Prospect Theory (Kahneman & Tversky, 1979) — loss salience creates a visible cost for ignoring warnings, designed to increase behavioral response. The −5/day rate and 3-day cap are calibrated design constants.

Gap Adjustment

```
bonus = min(catchUpDays × 2, 10) → added to FHS penalty = min(gapDays ×
3, 15) → subtracted from FHS
```

Basis: Ariely (2008) and Duhigg (2012) — consistency in tracking is more predictive of financial health than volume on a single day. The asymmetry (penalty > bonus) mirrors Prospect Theory's loss > gain finding. Values are calibrated design constants.

Lightweight Mode — Used when income is not set (students, informal economy)

Justified by UNSGSA (Gubbins, Mazzotta & Rhyne, 2021): transaction-derived indicators remain valid when full income data is unavailable.

Component	Formula	Basis
Spending Restraint (replaces Savings Rate)	`ratio = totalSpent ÷ selfSetLimit` Score = 25 × (1 − ((ratio − 0.80) ÷ 1.20)) [clamped 0–25] *No-limit fallback:* wantRatio = wantSpend ÷ totalSpend Score = 10 + 15 × (1 − ((wantRatio − 0.30) ÷ 0.40)) [floor: 10 pts]	Prospect Theory (Kahneman & Tversky, 1979) — want vs. need distinction as a behavioral health signal. 80% grace zone = personal finance buffer principle. Floor of 10 pts prevents penalizing users with no spending limit set.
Category Balance (replaces Budget Adherence)	`topRatio = topCategory ÷ discretionaryTotal` (Bills, Health, Education excluded) Score = 25 × (1 − ((topRatio − 0.40) ÷ 0.60)) [clamped 0–25] Full 25 pts when top category ≤ 40% of discretionary spend; 0 pts when one category = 100%	FinHealth Score® Spend pillar (Financial Health Network, 2021). Bills/Health/Education excluded because BSP CFIS (2025) confirms these are non-discretionary fixed obligations for Filipino households. 40% threshold is a calibrated design constant.
Logging Consistency (unchanged)	`25 × (loggedDays ÷ activeDays)`	Same as Full Mode — Deci & Ryan SDT (2000).

Habit Streak (added in Lightweight)	`25 × min(1, currentStreak ÷ 14)` Full 25 pts at 14-day consecutive streak	Deci & Ryan SDT (2000) Competence dimension; Lally et al. (2010) — consistent behavioral windows of ~14 days anchor initial habit formation routines.

Worked Example (Full Mode)

Sample user — 30-day period:
Income: ₱20,000 | Total spent: ₱15,000 → savingsRate = 25%
Active days: 20 | Hard overspend days: 2 | Soft overspend days: 3
Budget categories: 4 total, 3 on-budget | Logged days: 18 of 20

Component	Calculation	Score
Savings Rate (S)	25 × min(1, 0.25 ÷ 0.20) = 25 × min(1, 1.25) = 25 × 1.0	**25.0**
Overspend Control (O)	weighted = 2 + (3 × 0.5) = 3.5; ratio = 3.5 ÷ 20 = 0.175 25 × (1 − 0.175) = 25 × 0.825	**20.6**
Budget Adherence (B)	25 × (3 ÷ 4) = 25 × 0.75	**18.75**
Logging Consistency (L)	25 × (18 ÷ 20) = 25 × 0.90	**22.5**
Total (no adjustments triggered)		**86.85 ≈ 87 / 100**

Interpretation: A user who saves above target, has minor overspend on a few days, mostly follows budgets, and logs nearly every day earns an 87 — "Good" financial health status.

Note on Calibrated Constants

The following specific numeric values in the formula are **design calibrations** informed by the referenced theoretical frameworks but not directly derived from a single paper. They are prototype values subject to empirical adjustment in a longitudinal study:

Constant	Value	Theoretical Grounding
Warning Decay rate	−5 pts/day, max 3 days	Prospect Theory — loss salience
Gap bonus / penalty	+2 / −3 pts per day	Prospect Theory asymmetry; Ariely (2008); Duhigg (2012)

Soft overspend weight	0.5×	Prospect Theory / Nudge Theory — planned vs. habitual
Category Balance threshold	40% of discretionary	Concentration risk design principle
Spending Restraint grace zone	80% of limit	Personal finance buffer principle
Habit Streak target	14 days	SDT (Deci & Ryan, 2000); Lally et al. (2010)

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